

SHL Conversational Assessment Recommender

Architecture, Design Choices, Retrieval Pipeline & Evaluation

Yash Raghuvanshi | Research Intern, AI – Technical Design Document

1. System Overview

A conversational agent that understands hiring requirements through multi-turn dialogue, asks only necessary clarifying questions, retrieves relevant SHL assessments, refines recommendations as constraints change, and grounds every recommendation in the official SHL catalog – rather than letting the LLM generate assessment names directly, preventing hallucination.

Four components: Conversational Agent · Query Decomposer · Recommendation Engine · Semantic Retrieval System

2. Architecture

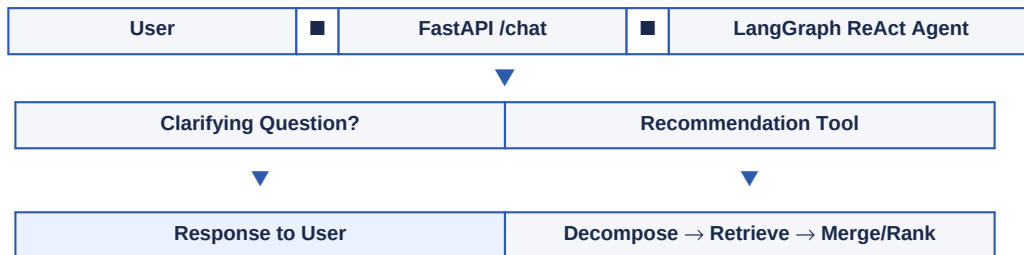


Figure 1. Request flow – agent branches between clarification and grounded retrieval.

3. Catalog Preprocessing

The SHL catalog (**377 assessments**) was converted into semantic documents combining name, description, job level, type, skills, duration, languages, and metadata. Each document was embedded offline with a sentence-transformer model and stored locally for fast similarity search at inference time.

4. Query Decomposition

Hiring requirements often bundle multiple independent needs (e.g. a Graduate Management Trainee battery covering cognitive, personality and situational judgement), which a single semantic search under-covers. An LLM-based decomposer splits such requests into multiple retrieval queries (“Cognitive assessment”, “Personality assessment”, “Situational judgement assessment”), each searched independently, improving coverage. Includes tolerant JSON parsing with fallback to the original query if decomposition fails.

5. Semantic Retrieval

For each retrieval query: embed → compute cosine similarity against all catalog embeddings → take top matches → rank by score. This tolerates wording that differs from catalog terminology (e.g. “Executive leadership benchmark” → Leadership reports, OPQ assessments, Executive scenario assessments).

6. Recommendation Engine

Results from all retrieval buckets are merged, deduplicated (keeping the highest similarity score per assessment), and ranked globally. Coverage metadata tracks which query produced each recommendation, aiding explainability.

7. Conversational Agent & Prompt Design

A LangGraph ReAct agent decides whether more information would materially change the recommendation (hiring vs. development, seniority, role family) before calling the recommendation tool, asking only the minimum clarifying questions. Conversation history lets follow-ups (“remove OPQ”, “add numerical reasoning”, “replace with a shorter assessment”) be treated as refinements rather than new scenarios.

Key prompt principles:

- Ask only materially-relevant clarifying questions
- Stop once information is sufficient
- Build a standalone request from history
- Never hallucinate assessment names
- Use only retrieved catalog assessments
- Treat follow-ups as refinements, not new scenarios
- Apply add/remove/replace conversationally
- Explain briefly, without exposing internal reasoning

8. Robustness

Safeguards: graceful fallback on decomposition failure · tolerant JSON parsing · hallucination prevention · off-topic refusal · semantic fallback when exact matches don't exist.

9. Evaluation

Dimension	Method
Retrieval Quality	Recall@10 over representative hiring queries
Recommendation Relevance	Expected vs. retrieved assessments
Groundedness	Every recommendation traced to the SHL catalog
Response Effectiveness	Latency, clarification quality, refinement handling

10. What Did Not Work

- Explicit state manager added complexity without improving conversation quality
- Excessive prompt instructions caused unnecessary clarifying questions
- Direct LLM-generated recommendations occasionally hallucinated
- Single-query retrieval missed multi-need hiring requests

11. Final Design Rationale

The LLM owns conversation management, clarification and refinement; the recommendation engine owns grounded retrieval from the SHL catalog. This separation keeps recommendations conversational, explainable, robust, and strictly grounded.